

Guide: Replacing an Orbit Prefix

Collatz proof project — September 5, 2026

Why merging helps

Imagine the smallest positive integer with an unbounded Collatz orbit. If a smaller positive integer eventually joins that orbit, its orbit is also unbounded, contradicting the choice of the smallest one. Lean proves this implication; guaranteeing that smaller predecessor remains open.

What the replacement certificate checks

A finite odd/even pattern determines an affine endpoint formula. Two patterns with the same length and number of odd steps have the same multiplier. Increasing the correction by $3^j\delta$ reduces the starting seed by δ while keeping the endpoint fixed. The proof checks that the new integer actually follows every specified parity. An arbitrary number in a plausible correction interval is insufficient.

The replacement is useful when $0 < \delta < n$, so its seed is positive and smaller. It need not grow at every intermediate step.

A working infinite family

Any initial run of odd steps followed by two even steps can be replaced by a pattern starting from the integer immediately below the original seed. For example, 15 and 14 both reach 20 after six shortcut steps, with four odd steps each. This classical Garner family is now formalized here; it is not a new discovery or a universal coverage result.

Why this version cannot cover every seed

Lean proves that 27 never merges with a smaller positive seed at the same time with the same number of odd steps. The proof checks the initial 70 steps, then uses the repeating cycle: 27 has accumulated too many odd steps for any smaller seed to catch up under these two restrictions.

Since 27 reaches one, this refutes unrestricted universal coverage, but not coverage conditional on least-unboundedness.

Different odd counts: a bound and a test

Lean proves that a merging seed with fewer odd steps must exceed $2n$ when $n \geq 2^t$. It therefore cannot be the smaller predecessor we need. More odd steps introduce a congruence condition modulo a power of three. The experiment accounts for every nonnegative quotient in each tested binary cylinder, including larger seeds than its canonical representative.

At depth 18, equal-count replacement covers 1,391 of 7,495 patterns whose every prefix coefficient is at least one. Varying the odd count adds no coverage among the remaining 6,104 cylinders, for any quotient. The same zero-addition result holds through depth 17. These are finite-depth Python results, not a theorem for arbitrary lengths or evidence of divergence.

Reproduce and continue

From `lean/`, run `lake build Collatz.WordSurgery`. The new experiment is `analysis/variable_count_surgery.py`; its derivation and tests are in matching `analysis/` files. Different word lengths or odd counts may help, but neither nondivergence nor exclusion of all nontrivial cycles has been proved here.